Contents

Preface ix

1 Probability and Distributions 1
- 1.1 Introduction . . . 1
- 1.2 Set Theory . . . 3
- 1.3 The Probability Set Function . . . 10
- 1.4 Conditional Probability and Independence . . . 21
- 1.5 Random Variables . . . 32
- 1.6 Discrete Random Variables . . . 40
  - 1.6.1 Transformations . . . 42
- 1.7 Continuous Random Variables . . . 44
  - 1.7.1 Transformations . . . 46
- 1.8 Expectation of a Random Variable . . . 52
- 1.9 Some Special Expectations . . . 57
- 1.10 Important Inequalities . . . 68

2 Multivariate Distributions 73
- 2.1 Distributions of Two Random Variables . . . 73
  - 2.1.1 Expectation . . . 79
- 2.2 Transformations: Bivariate Random Variables . . . 84
- 2.3 Conditional Distributions and Expectations . . . 94
- 2.4 The Correlation Coefficient . . . 102
- 2.5 Independent Random Variables . . . 110
- 2.6 Extension to Several Random Variables . . . 117
  - 2.6.1 *Multivariate Variance-Covariance Matrix . . . 123
- 2.7 Transformations for Several Random Variables . . . 126
- 2.8 Linear Combinations of Random Variables . . . 134

3 Some Special Distributions 139
- 3.1 The Binomial and Related Distributions . . . 139
- 3.2 The Poisson Distribution . . . 150
- 3.3 The Γ, χ^2, and β Distributions . . . 156
- 3.4 The Normal Distribution . . . 168
  - 3.4.1 Contaminated Normals . . . 174

3.5 The Multivariate Normal Distribution 178
3.5.1 *Applications 185
3.6 t- and F-Distributions 189
3.6.1 The t-distribution 189
3.6.2 The F-distribution 191
3.6.3 Student's Theorem 193
3.7 Mixture Distributions 197

4 Some Elementary Statistical Inferences 203
4.1 Sampling and Statistics 203
4.1.1 Histogram Estimates of pmfs and pdfs 207
4.2 Confidence Intervals 214
4.2.1 Confidence Intervals for Difference in Means 217
4.2.2 Confidence Interval for Difference in Proportions 219
4.3 Confidence Intervals for Parameters of Discrete Distributions 223
4.4 Order Statistics 227
4.4.1 Quantiles 231
4.4.2 Confidence Intervals for Quantiles 234
4.5 Introduction to Hypothesis Testing 240
4.6 Additional Comments About Statistical Tests 248
4.7 Chi-Square Tests 254
4.8 The Method of Monte Carlo 261
4.8.1 Accept–Reject Generation Algorithm 268
4.9 Bootstrap Procedures 273
4.9.1 Percentile Bootstrap Confidence Intervals 273
4.9.2 Bootstrap Testing Procedures 276
4.10 *Tolerance Limits for Distributions 284

5 Consistency and Limiting Distributions 289
5.1 Convergence in Probability 289
5.2 Convergence in Distribution 294
5.2.1 Bounded in Probability 300
5.2.2 Δ-Method 301
5.2.3 Moment Generating Function Technique 303
5.3 Central Limit Theorem 307
5.4 *Extensions to Multivariate Distributions 314

6 Maximum Likelihood Methods 321
6.1 Maximum Likelihood Estimation 321
6.2 Rao–Cramér Lower Bound and Efficiency 327
6.3 Maximum Likelihood Tests 341
6.4 Multiparameter Case: Estimation 350
6.5 Multiparameter Case: Testing 359
6.6 The EM Algorithm 367

7 Sufficiency **375**
7.1 Measures of Quality of Estimators . . . 375
7.2 A Sufficient Statistic for a Parameter . . . 381
7.3 Properties of a Sufficient Statistic . . . 388
7.4 Completeness and Uniqueness . . . 392
7.5 The Exponential Class of Distributions . . . 397
7.6 Functions of a Parameter . . . 402
7.7 The Case of Several Parameters . . . 407
7.8 Minimal Sufficiency and Ancillary Statistics . . . 415
7.9 Sufficiency, Completeness, and Independence . . . 421

8 Optimal Tests of Hypotheses **429**
8.1 Most Powerful Tests . . . 429
8.2 Uniformly Most Powerful Tests . . . 439
8.3 Likelihood Ratio Tests . . . 447
8.4 The Sequential Probability Ratio Test . . . 459
8.5 Minimax and Classification Procedures . . . 466
8.5.1 Minimax Procedures . . . 466
8.5.2 Classification . . . 469

9 Inferences About Normal Models **473**
9.1 Quadratic Forms . . . 473
9.2 One-Way ANOVA . . . 478
9.3 Noncentral χ^2 and F-Distributions . . . 484
9.4 Multiple Comparisons . . . 486
9.5 The Analysis of Variance . . . 490
9.6 A Regression Problem . . . 497
9.7 A Test of Independence . . . 506
9.8 The Distributions of Certain Quadratic Forms . . . 509
9.9 The Independence of Certain Quadratic Forms . . . 516

10 Nonparametric and Robust Statistics **525**
10.1 Location Models . . . 525
10.2 Sample Median and the Sign Test . . . 528
10.2.1 Asymptotic Relative Efficiency . . . 533
10.2.2 Estimating Equations Based on the Sign Test . . . 538
10.2.3 Confidence Interval for the Median . . . 539
10.3 Signed-Rank Wilcoxon . . . 541
10.3.1 Asymptotic Relative Efficiency . . . 546
10.3.2 Estimating Equations Based on Signed-Rank Wilcoxon . . . 549
10.3.3 Confidence Interval for the Median . . . 549
10.4 Mann–Whitney–Wilcoxon Procedure . . . 551
10.4.1 Asymptotic Relative Efficiency . . . 555
10.4.2 Estimating Equations Based on the Mann–Whitney–Wilcoxon 556
10.4.3 Confidence Interval for the Shift Parameter Δ . . . 557
10.5 General Rank Scores . . . 559